ASEN 6044

Homework #2

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Problem 1:

We are given a prior and measurment likelihood given by

$$P(\theta) = \begin{cases} \alpha e^{-\alpha\theta} & \theta \geq 0 \\ 0 \quad \theta < 0 \end{cases}$$

$$P(y \mid \theta) = \frac{\theta^y e^{-\theta}}{y!} \quad y \in \mathbb{Z}^+$$

From Bayes Rule, we know the posterior distribution is given by

$$P(\theta \mid y) = \frac{P(y \mid \theta)P(\theta)}{P(y)}$$

Where $P(y)$ is given by the marginalization of the joint distribution $P(\theta, y)$, that is

$$P(y) = \int_{-\infty}^{\infty} P(\theta, y)d\theta = \int_{-\infty}^{\infty} P(y \mid \theta)P(\theta)d\theta = \int_{0}^{\infty} \alpha e^{-\alpha\theta}\frac{\theta^y e^{-\theta}}{y!}d\theta$$

Since we are given y, $P(y)$ is a constant and is not dependant on θ. Thus, our posterior distribution will be proportional to the measurment likelihood times the prior.

$$P(\theta \mid y) \propto \alpha e^{-\alpha\theta}\frac{\theta^y e^{-\theta}}{y!} = \frac{\alpha\theta^y e^{-\theta(1+\alpha)}}{y!}$$

Recall the pdf of a gamma distribution:

$$f(x) = \Gamma(\gamma, \beta) = \frac{\beta^\gamma x^{\gamma-1}e^{-\beta x}}{\Gamma(\gamma)}$$

Where $\Gamma(\gamma)$ is the gamma function. For positive integers, the gamma function returns $\Gamma(n) = (n-1)!$. Since y is a positive integer, it's clear that our posterior distribution is a gamma distribution with parameters $\gamma = y + 1$ and $\beta = \alpha + 1$.

a) The MMSE estimate for our posterior distribution is the expected value of the distribution.

$$\hat{\theta}_{MMSE} = \mathbb{E}[P(\theta \mid y)]$$

For a gamma distribution, the expected value is given by $\mathbb{E}[\Gamma(\gamma, \beta)] = \frac{\gamma}{\beta}$. Thus, the MMSE estimate for our posterior distribution is

$$\boxed{\hat{\theta}_{MMSE} = \frac{y+1}{\alpha+1}}$$

b) The MAP estimate for our posterior distribution is the maximum value of our posterior. We can find the maximum by taking the derivative of our posterior with respect to θ and setting equal to 0.

$$\frac{d}{d\theta}\Big[\frac{\alpha\theta^y e^{-\theta(1+\alpha)}}{y!}\Big] = \frac{\alpha[y\theta^{y-1}e^{-\theta(1+\alpha)} - (1+\alpha)e^{-\theta(1+\alpha)}\theta^y]}{y!} = 0$$
$$[y\theta^{y-1} - (\alpha+1)\theta^y]e^{-\theta(1+\alpha)} = 0 \rightarrow (\alpha+1)\theta^y = y\theta^{y-1}$$
$$\alpha + 1 = y\theta^{-1} \rightarrow \boxed{\hat{\theta}_{MAP} = \frac{y}{\alpha+1}}$$

Problem 2:

We are given that a continous random variable θ is uniformly distributed on the interval $[0,1]$, adn that measurments are given by $y = \theta + n$ where n's distribution is given as

$$P(n) = \begin{cases} e^{-n} & n \geq 0 \\ 0 & n < 0 \end{cases}$$

From the following pdf, we can derive our measurement likelihood $P(y \mid \theta)$. Since our noise is additive, we have $n = y - \theta$. Plugging this into $P(n)$ we have

$$P(y \mid \theta) = P(y - \theta) = \begin{cases} e^{-y+\theta} & y \geq \theta \\ 0 & y < \theta \end{cases}$$

For a fixed observation y, this is identical to the measurment likelihood, $P(y \mid \theta)$. Applying Bayes Rule, we can derive an expression for our posterior.

$$P(\theta \mid y) = \frac{P(y \mid \theta)P(\theta)}{\int_{-\infty}^{\infty} P(y \mid \theta)P(\theta)d\theta}$$

The bounds on the posterior must be consistent with the constraints placed on the prior and likelihood. We know $\theta \in [0,1]$ and $y \geq \theta$. Thus, we know our posterior estimate should be cut off at $\theta = 1$ is $y \geq 1$. However if $y < 1$, we know that theta cannot be larger than y. Thus an upperbound on our posterior distribution becomes $u = \min(1, y)$. We can use this upper bound to solve for the normalizing integral in posterior expression.

$$\int_{-\infty}^{\infty} P(y \mid \theta)P(\theta)d\theta = \int_0^u P(y \mid \theta)P(\theta)d\theta = \int_0^u e^{-y}e^{\theta}d\theta$$
$$= e^{-y}[e^u - 1]$$

And out posterior becomes

$$P(\theta \mid y) = \begin{cases} \frac{e^\theta}{e^u - 1} & 0 \leq \theta \leq u \\ 0 & \text{o.w} \end{cases} \qquad u = \min(1, y)$$

a) $\hat{\theta}_{MMSE} = \mathbb{E}[P(\theta \mid y)]$ by definition. Thus we can evaluate the expected value to recieve an expression for $\hat{\theta}_{MMSE}$.

$$\mathbb{E}[P(\theta \mid y)] = \int_0^u \theta \frac{e^\theta}{e^u - 1} d\theta = \frac{1}{e^u - 1}\Big[\theta e^\theta\Big|_0^u - \int_0^u e^\theta d\theta\Big]$$
$$= \frac{1}{e^u - 1}\Big[ue^u - e^u + 1\Big] = \frac{ue^u - e^u + 1}{e^u - 1}$$

Thus $\hat{\theta}_{MMSE} = \mathbb{E}[P(\theta \mid y)] = \boxed{\dfrac{ue^u - e^u + 1}{e^u - 1}}$

b) By taking the derivative of our posterior we can derive an expression for $\hat{\theta}_{MAP}$.

$$\frac{d}{d\theta}\left[\frac{e^\theta}{e^u - 1}\right] = \frac{e^\theta}{e^u - 1}$$

Since this function is monotonic increasing, it's maximum must lie a the end of the interval. Thus, $\boxed{\hat{\theta}_{MAP} = \min(u, 1)}$.

Problem 3:

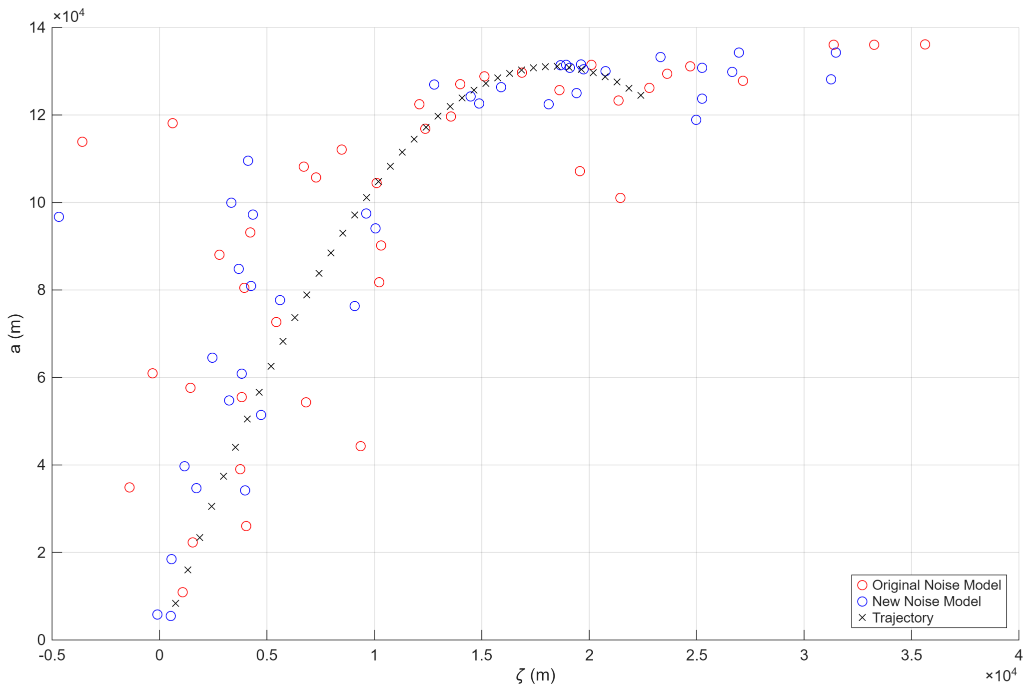

Figure 1: Comparison of Old and New Noisy Measurments

a) From Figure 1 we see generally that the measurments from the new noisy range model draw out a similar shape to that of the old range model. Without any prior knowledge regarding the problem, it even appears that both sets of measurments have the same noise model.

b) TODO: Best x_0, Variance of range? Is it even supposed to work??

c) The negative log-likelihood function, $\mathcal{L}(x_0; Y_{1:N}) = J_{NLL}$ is defined as the negative logarithm of the measurment likelihood functions, that is

$$J_{NLL} = -\log(P(Y_{1:N} \mid x_0))$$

To derive an expression for $P(Y_{1:N} \mid x_0)$, we look at the given measurment model:

$$\rho_k = \sqrt{(L-\xi(t_k))^2 + (a(t_k))^2} + v_{r,k} \quad v_{r,k} \sim \mathcal{T}_{v_{r,k}}\Big(0, \nu = \frac{1}{2}\Big) \tag{1}$$

$$\theta_k = \arctan\bigg(\frac{a(t_k)}{L-\xi(t_k)}\bigg) + v_{b,k} \quad v_{b,k} \sim \mathcal{N}_{v_{b,k}}\Big(0, \sigma_b^2 = 0.005\Big) \tag{2}$$

Where $v_{r,k}$ and $v_{b,k}$ are the range and bearing noise respectively. To derive the measurment likelihood $P(Y_{1:N} \mid x_0)$ we assume that the measurments are conditionally independent given x_0, that is $P(Y_{1:N} \mid x_0) = \prod_{k=1}^{N} P(y_k \mid x_0)$. We also assume that the range and bearing measurments are independent of each other, our expression further simplifies to the following:

$$P(Y_{1:N} \mid x_0) = \prod_{k=1}^{N} P(\rho_k \mid x_0) P(\theta_k \mid x_0) \tag{3}$$

Since our noise is additive, solving for the respective probabilities is fairly straightforward. From (1) we can derive $P(\rho_k \mid x_0)$ as follows:

$$v_{r,k} = \rho_k - \sqrt{(L-\xi(t_k))^2 + (a(t_k))^2}$$
$$P(v_{r,k}) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}}\Big(1 + \frac{v_{r,k}^2}{\nu}\Big)^{-\frac{\nu+1}{2}}$$

Substituting for $v_{r,k}$ we get the following expression in terms of our initial conditions.

$$P(\rho_k \mid x_0) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}}\Big(1 + \frac{(\rho_k - \sqrt{(L-\xi_0-\dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 t_k - 0.5 g t_k^2)})}{\nu}\Big)^{-\frac{\nu+1}{2}} \tag{4}$$

We can follow a similar process to derive $P(\theta_k \mid x_0)$.

$$v_{b,k} = \theta_k - \arctan\bigg(\frac{a(t_k)}{\xi(t_k)}\bigg)$$
$$P(v_{b,k}) = \frac{1}{\sigma_b\sqrt{2\pi}} \exp\bigg\{\frac{-1}{2\sigma_b^2} v_{b,k}^2\bigg\}$$

$$P(\theta_k \mid x_0) = \frac{1}{\sigma_b\sqrt{2\pi}} \exp\left\{\frac{-\Big[\theta_k - \arctan\Big(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\Big)\Big]^2}{2\sigma_b^2}\right\} \tag{5}$$

Substituting (5) and (4) into (3) and taking the negative logarithm we recieve the following expression for our negative log likelihood function:

$$
\begin{aligned}
\mathcal{L}(x_0; Y_{1:N}) &= -\sum_{k=1}^{N} \log[P(\rho_k \mid x_0)P(\theta_k \mid x_0)] \\
&= -\sum_{k=1}^{N} \log\left(\frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}}\frac{1}{\sigma_b\sqrt{2\pi}}\right) \\
&\quad -\sum_{k=1}^{N} -\frac{\nu+1}{2}\log\left(1+\frac{(\rho_k - \sqrt{(L-\xi_0-\dot{\xi}_0 t_k)^2 + (a_0+\dot{a_0}t_k - 0.5gt_k^2)})}{\nu}\right) \\
&\quad -\sum_{k=1}^{N} \frac{-\left[\theta_k - \arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi}_0 t_k}\right)\right]^2}{2\sigma_b^2}
\end{aligned} \tag{6}
$$

Both the NLL and NLS cost functions seek to find an x_0 that minimizes the cost function given observations $Y_{1:N}$, however they each approach the problem in a different way. NLS is defined to minimize the sum of the squared residuals, while maximum likelihood tries to find the x_0 which makes 'the most sense' given all the observations. Both assume the measurment noise characteristics are known, NLS uses a least squares cost function, meaning we only consider the 1st/2nd order moments. As we have seen with the Student's-t distribution, this does not accuratly capture higher order moments. Maximum likelihood is robust to non-Gaussian sensor noise unlike the least-squares cost function we have selected for NLS.

d) In order to minimize (6), we take the gradient with respect to x_0, or $\nabla_{x_0}\mathcal{L}(x_0; Y_{1:N})$, where

$$
\nabla_{x_0} = \begin{bmatrix} \frac{\partial}{\partial \xi_0} \\ \frac{\partial}{\partial \dot{\xi}_0} \\ \frac{\partial}{\partial a_0} \\ \frac{\partial}{\partial \dot{a_0}} \end{bmatrix}
$$

$$
\nabla_{x_0}\mathcal{L}(x_0; Y_{1:N}) = -\nabla_{x_0}\sum_{k=1}^{N} \log[P(\rho_k \mid x_0)P(\theta_k \mid x_0)]
$$

It's easy to show that the expression simplifies to the following, where the constant terms evaluate to zero.

$$
\begin{aligned}
&= -\nabla_{x_0}\left[\frac{-\nu+1}{2}\log\left(1+\frac{\left(\rho_k - \sqrt{(L-\xi_0-\dot{\xi}_0 t_k)^2 + (a_0+\dot{a_0} - 0.5gt_k^2)^2}\,\right)^2}{\nu}\right)\right] \\
&\quad - \nabla_{x_0}\left[\frac{-1}{2}\left[\frac{\theta_k - \arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi}_0 t_k}\right)}{\sigma_b}\right]^2\right]
\end{aligned} \tag{7}
$$

Equation (7) consists of two gradient terms to evaluate. Starting with the first term, we can take each partial derivative independently and stack the result into a vector:

$$\nabla_{x_0}\left[\frac{-\nu+1}{2}\log\left(1+\frac{\left(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0}t_k)^2+(a_0+\dot{a_0}-0.5gt_k^2)^2}\;\right)^2}{\nu}\right)\right]$$

$$\frac{\partial}{\partial\xi_0}\left[\frac{-\nu+1}{2}\log\left(1+\frac{\left(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0}t_k)^2+(a_0+\dot{a_0}-0.5gt_k^2)^2}\;\right)^2}{\nu}\right)\right]$$
$$=\frac{2(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(L-\xi_0-\dot{\xi_0}t)}{(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}}$$

$$\frac{\partial}{\partial\dot{\xi_0}}\left[\frac{-\nu+1}{2}\log\left(1+\frac{\left(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0}t_k)^2+(a_0+\dot{a_0}-0.5gt_k^2)^2}\;\right)^2}{\nu}\right)\right]$$
$$=\frac{2t(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(L-\xi_0-\dot{\xi_0}t)}{(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}}$$

$$\frac{\partial}{\partial a_0}\left[\frac{-\nu+1}{2}\log\left(1+\frac{\left(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0}t_k)^2+(a_0+\dot{a_0}-0.5gt_k^2)^2}\;\right)^2}{\nu}\right)\right]$$
$$=\frac{-2(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(a_0+\dot{a_0}t-0.5gt^2)}{(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}}$$

$$\frac{\partial}{\partial\dot{a_0}}\left[\frac{-\nu+1}{2}\log\left(1+\frac{\left(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0}t_k)^2+(a_0+\dot{a_0}-0.5gt_k^2)^2}\;\right)^2}{\nu}\right)\right]$$
$$=\frac{-2t(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(a_0+\dot{a_0}t-0.5gt^2)}{(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}}$$

Therefore,

$$\nabla_{x_0}\left[\frac{-\nu+1}{2}\log\left(1+\frac{\left(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0}t_k)^2+(a_0+\dot{a_0}-0.5gt_k^2)^2}\;\right)^2}{\nu}\right)\right]$$

$$
= \frac{-\nu+1}{2}\begin{bmatrix}
\frac{2(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(L-\xi_0-\dot{\xi_0}t)}{\left(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu\right)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}} \\
\frac{2t(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(L-\xi_0-\dot{\xi_0}t)}{\left(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu\right)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}} \\
\frac{-2(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(a_0+\dot{a_0}t-0.5gt^2)}{\left(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu\right)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}} \\
\frac{-2t(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(a_0+\dot{a_0}t-0.5gt^2)}{\left(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu\right)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}}
\end{bmatrix} \tag{8}
$$

We will follow a similar process for the second gradient term.

$$
\nabla_{x_0}\Big[\frac{-1}{2}\Big[\frac{\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)}{\sigma_b}\Big]^2\Big] = \frac{-1}{2\sigma_b^2}\nabla_{x_0}\Big[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)\Big]^2
$$

$$
\frac{\partial}{\partial\xi_0}\Big[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)\Big]^2 = \frac{-2t[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)](a_o+\dot{a_0}t_k-0.5gt_k^2)}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)^2}
$$

$$
\frac{\partial}{\partial\dot{\xi_0}}\Big[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)\Big]^2 = \frac{-2[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)](a_o+\dot{a_0}t_k-0.5gt_k^2)}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)^2}
$$

$$
\frac{\partial}{\partial a_0}\Big[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)\Big]^2 = \frac{-2[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)]}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)}
$$

$$
\frac{\partial}{\partial\dot{a_0}}\Big[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)\Big]^2 = \frac{-2t[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)]}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)}
$$

Therefore,

$$\nabla_{x_0}\left[\frac{-1}{2}\left[\frac{\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)}{\sigma_b}\right]^2\right]=\frac{-1}{2\sigma_b^2}\begin{bmatrix}\frac{-2[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)](a_o+\dot{a_0}t_k-0.5gt_k^2)}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)^2}\\ \frac{-2t[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)](a_o+\dot{a_0}t_k-0.5gt_k^2)}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)^2}\\ \frac{-2[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)]}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)}\\ \frac{-2t[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)]}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)}\end{bmatrix} \tag{9}$$

Finally, we can substitute equations (8) and (9) into equation (7) and we are left with an expression for the gradient vector of $\mathcal{L}_{NLL}$

$$\begin{aligned}\nabla_{x_0}\mathcal{L}(x_0;Y_{1:N})=\sum_{k=1}^{N}\Bigg[\frac{\nu-1}{2}&\begin{bmatrix}\frac{2(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0}t)^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(L-\xi_0-\dot{\xi_0}t)}{\left(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu\right)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}}\\ \frac{2t(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0}t)^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(L-\xi_0-\dot{\xi_0}t)}{\left(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu\right)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}}\\ \frac{-2(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0}t)^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(a_0+\dot{a_0}t-0.5gt^2)}{\left(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu\right)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}}\\ \frac{-2t(\rho_k-\sqrt{(L-\xi_0-\dot{\xi_0}t)^2+(a_0+\dot{a_0}t-0.5gt_k^2)^2})(a_0+\dot{a_0}t-0.5gt^2)}{\left(\left(\rho-\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}\right)^2+\nu\right)\sqrt{(L-\xi_0-\dot{\xi_0})^2+(a_o+\dot{a_0}t-0.5gt^2)^2}}\end{bmatrix}\\ +\frac{1}{2\sigma_b^2}&\begin{bmatrix}\frac{-2[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)](a_o+\dot{a_0}t_k-0.5gt_k^2)}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)^2}\\ \frac{-2t[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)](a_o+\dot{a_0}t_k-0.5gt_k^2)}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)^2}\\ \frac{-2[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)]}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)}\\ \frac{-2t[\theta_k-\arctan\left(\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}\right)]}{([\frac{a_0+\dot{a_0}t_k-0.5gt_k^2}{L-\xi_0-\dot{\xi_0}t_k}]^2+1)(L-\xi_0-\dot{\xi_0}t_k)}\end{bmatrix}\Bigg]\end{aligned} \tag{10}$$

e)

f)

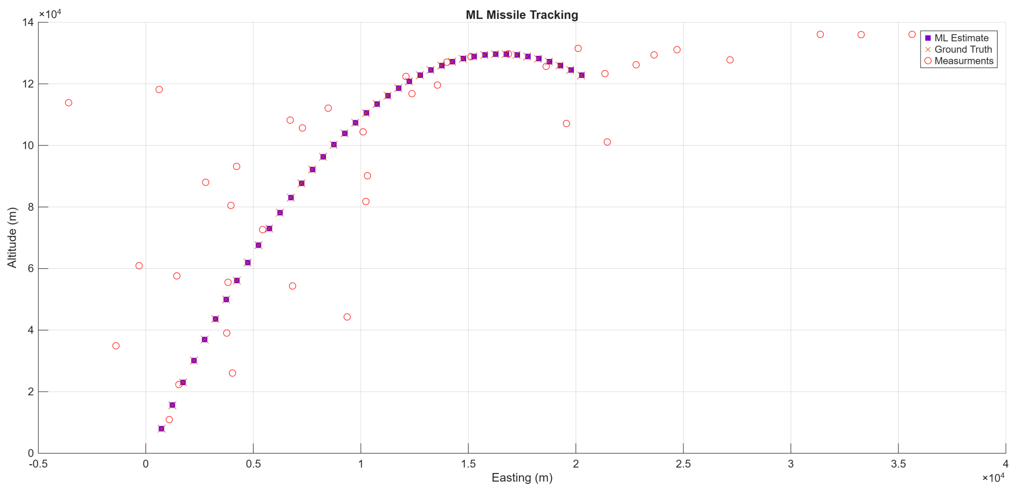

Figure 2: Maximum Likelihood Estimated Trajectory